INCH-POUND

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9 May 1994
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MILITARY SPECIFICATION

CYLINDER, RECUPERATOR, RECOIL, 105mm HOWITZER

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Cylinder, Recuperator, Recoil, which is used on the 105mm, Light Towed Howitzer, M119A1 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

BB-N-411 - Nitrogen, Technical

MILITARY

MIL-H-5606 - Hydraulic Fluid, Petroleum Base
MIL-H-6083 - Hydraulic Fluid, Petroleum Base for Preservation and Operation

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC 1015**

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MIL-H-6088 - Heat Treatment of Aluminum Alloy
MIL-H-6875 - Heat Treatment of Steel, Process for
MIL-S-7742 - Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-I-45607 - Inspection Equipment, Aquisition, Maintenance and Disposition of
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-W-63150 - Weapons and Support Materiel Standard

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services
FED-STD-595 - Colors

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-130 - Identification Marking of U.S. Military Property
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification
MIL-STD-453 - Inspection, Radiographic
MIL-STD-1252 - Inertia, Friction Welding Process, Procedure and Performance Qualification
MIL-STD-1261 - Arc Welding Procedures for Construction Steels
MIL-STD-1264 - Radiographic Inspection for Soundness of Welds in Steel by Comparison to Graded ASTM E390 Reference Radiographs
MIL-STD-2073-1 - DoD Materiel Procedures for Development and Application of Packaging
MIL-STD-6866 - Inspection, Liquid Penetrant
MIL-STD-45662 - Calibration Systems Requirements

(unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 40, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications

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form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

A9347218 - Finish Methods for Metal Surfaces
F12938299 - Recuperator

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture
ANSI Y14.5 - Dimensioning and Tolerancing

(Application for copies should be addressed to The American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York 10017)

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 - Symbols for Welding and Nondestructive Testing
AWSC5.5-80 - Gas, Tungsten Arc Welding, Recommended Practices for

(Application for copies should be addressed to The American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E1444 - Standard Practice for Magnetic Particle Examination

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained. (See contract provisions for additional precedence criteria.)

3. REQUIREMENTS

3.1 Materials. The recuperator and resevoir assembly shall conform to the materials, construction and design requirements specified herein and on Drawing 12938299.

3.1.1 Heat treatment. Unless otherwise specified on the applicable drawings, heat treatment for aluminum alloys shall conform to MIL-H-6088 and for steel shall conform to MIL-H-6875.

3.1.2 Dimensions. Dimensions and tolerances of parts and assemblies of the recuperator and resevoir assembly shall conform to the detailed drawings of Drawing 12938299. Interpretation of the dimensions and tolerances shall be in accordance with ANSI Y14.5.

3.1.3 Surface texture. Surface texture shall be as specified on the applicable drawings. Interpretation of surface roughness symbols shall be as defined in ANSI B46.1.

3.1.4 Protective finishes.

3.1.4.1 Inorganic finishes. Parts requiring an inorganicfinish shall be finished or coated as indicated on the applicable drawings and in accordance with MIL-STD-171 and Drawing A9347218.

3.1.4.2 Painting. The parts and assemblies shall be painted as specified on the drawing and in accordance with MIL-STD-171 and Drawing A9347218. The visible surfaces of attaching hardware (nuts, screws, washers, etc), exposed unpainted surfaces of mounting faces, or blemished areas shall be primed and painted to match surrounding surfaces after final assembly. To avoid the necessity of painting exposed surfaces of cadmium plated hardware (including nuts, screws, washers, etc), the hardware may be dyed with a supplementary chromate finish, Forest Green, equivalent to color 34904 of FED-STD-595.

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3.1.5 Welding. Interpretation of weld symbols specified on the applicable drawings shall be as defined in AWS A2.4. Actual production of weldments shall not proceed prior to Government approval of the contractor's welding procedures and test specimens of the applicable welding specifications. Qualification of arc welding procedures and welders shall be in accordance with MIL-STD-248. All plug welds shall be flush and free of snags or protrusions. Grinding flush is not required unless specified on the drawing or in the welding specification. Weld splatter and similar excess weld shall be removed.

3.1.5.1 Arc welding, ferrous metals. Welding of steel parts and assemblies shall be in accordance with weld classification of MIL-STD-1261 referenced on applicable drawings and AWS C5.5-80.

3.1.5.2 Welding, inertia. Inertia friction welding shall be in accordance with MIL-STD-1252.

3.1.5.3 Weld overlay. A First Article Test plan and a process control plan for weld overlay procedures called out on part drawings shall be prepared by the contractor and submitted to the Government for approval. The weld overlay process shall be in accordance with MIL-STD-248.

3.1.5.4 Magnetic particle inspection. When magnetic particle inspection is indicated on the drawing, it shall be in accordance with ASTM E1444.

3.1.5.5 Penetrant inspection. When penetrant inspection is indicated on the drawing, it shall be in accordance with MIL-STD-6866.

3.1.5.6 Radiographic inspection. Radiographic inspection shall be in accordance with MIL-STD-453. Steel weldments shall be inspected to the standards of MIL-STD-1264 and as referenced on the applicable drawings.

3.1.6 Threads. Threads shall conform to the thread series and class indicated on the applicable drawing, and be in accordance with FED-STD-H28. Unless otherwise specified, coated or plated class 3 threads shall conform to the coated thread data specified in MIL-S-7742.

3.1.6.1 Torque, threaded fasteners. Unless otherwise specified on the applicable drawing, threaded fasteners S.A.E. Grade 5, shall be torqued in accordance with the following torque requirements:

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Thread Size	Torque
No. 4-48UNF	9-10 in-lb
No. 6-32UNC	16-18 in-lb
No. 8-32UNC	30-33 in-lb
No. 10-24UNC	40-45 in-lb
No. 10-32UNF	45-50 in-lb
1/4-20UNC	96-120 in-lb
5/16-18UNC	11-18 ft-lb
3/8-16UNC	30-33 ft-lb
1/2-13UNC	75-80 ft-lb
5/8-11UNC	150-165 ft-lb
3/4-16UNF	300-330 ft-lb

This chart is based on the use of clean, dry threads. Other conditions apply as follows:

a. Reduce torque by 10 percent when oil is used as lubricant

b. Reduce torque by 20 percent when new fasteners are used.

c. Reduce torque by 30 percent when threading fasteners into aluminum.

3.2 First article. A sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.3 Performance characteristics.

3.3.1 General. The hydraulic portion of the recuperator shall be filled with hydraulic fluid, bled and refilled until all air and measurable moisture are eliminated. The initial and subsequent filling of the hydraulic portion shall be with hydraulic oil conforming to MIL-H-5606 or MIL-H-6083 and filtered through a 5 micron filter. The pneumatic system of the recuperator shall be filled with nitrogen conforming to BB-N-411 (Type I, Grade B, Class I). The completed recuperator and resevoir assembly shall conform to the characteristic specified on Drawing F12938299 and the requirements of 3.3.2 through 3.3.6.

3.3.2 Gymnastication. The recuperator and resevoir shall be subjected to the gymnastication test specified in 4.5.1. The assembly shall be extended to a length of 44+/-0.5 inches and shall

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return to its rest position with out slamming. This shall be accomplished prior to functional firing.

3.3.2.1 Damage to sliding surfaces. The recuperator rod, the Outside Diameter of the floating cylinder, the Inside Diameter of the cylinder and the Inside Diameter of the recuperator body cylinder shall show no evidence of damage after gymnastication.

3.3.3 Functional firing. The recuperator and resevoir assembly, charged with nitrogen to a pressure of 750+/-50 psi, shall withstand functional firing as specified in Table I without evidence of malfunction or failure. Operation during the complete recoil cycle shall be smooth and continuous within the limits established herein. Conditions such as erratic motion, surge, hang-up during recoil or counter-recoil action shall not be permitted. Cycle times shall not exceed 4 seconds with the varo valve set at 16+/-4 clicks.

Table I
Functional Firing

Round Number 1/	Prop Charge	Zone	Quadrant Elevation (mil)	Recoil Length (inches) 2/
warmer	M67	optional	90	-
1 3/	M67	7	1244	12/16
2	M200	8	300	42/44.5
3 4/	M200	8	800	17/21
4	M67	7	90	38/43

1/ Standard 105mm projectile inert loaded to 30.85 ± 0.35 pounds.

2/ Recoil lengths are for information only.

3/ Firing sequence is optional. If there is a delay of more than 10 minutes between rounds, an M67 warmer must be fired.

4/ Charges will be heated to +145 degrees Fahrenheit.

3.3.3.1 Damage to sliding surfaces. The recuperator rod, the Outside Diameter of the floating cylinder, the Inside Diameter of the cylinder and the Inside Diameter of the recuperator body cylinder shall show no evidence of damage after functional firing.

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3.3.4 Contaminated/ aerated oil.

3.3.4.1 Contaminated oil. Immediately after functional firing, a two ounce sample of oil shall be removed from the oil fill port in the recuperator body into a straight sided glass beaker. Metallic and nonmetallic particle content shall not exceed 400 microns in size.

3.3.4.2 Aerated oil. After functional firing, oil withdrawn from the recuperator into a straight sided beaker shall be in accordance with the following: Within the first minute after withdrawal, the oil shall be five percent or less foam by volume. When exposed to the atmosphere for fifteen minutes, the aeration bubbles shall disappear.

3.3.5 Leakage.

3.3.5.1 Nitrogen leakage. There shall be no evidence of nitrogen leakage from a newly manufactured recuperator assembly.

3.3.5.2 Oil leakage. Oil leakage shall not exceed 15 drops per minute in a newly manufactured recuperator.

3.3.6 Operating temperature range. The recuperator assembly shall function in the temperature range of -50 degrees Fahrenheit to +145 degrees Fahrenheit.

3.4 Reliability, maintainability and durability. The recuperator assembly manufactured in accordance with this specification shall be free from any defects that compromise, limit or reduce the end item howitzer or component capability. The recuperator assembly shall be subject to examination and test by the Government to prove capability and performance in accordance with the following:

3.4.1 Durability. The durability requirement is 3500 Equivalent Full Charge (EFC) rounds without rebuild or overhaul, where at least 2000 of the required rounds are fired with the M200 propelling charge.

3.4.2 Seal durability. The dynamic seals of the recuperator shall demonstrate a durability of 6000 operational mode summary rounds.

3.5 Identification and instructional marking. Identification and instructional markings shall be applied in accordance with

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F12938299. Other marking, including application of part numbers and instructional marking, shall be as specified on the applicable drawings in accordance with MIL-STD-130. Serial numbers shall be furnished by the procuring activity (see 6.2).

3.6 Workmanship. Workmanship and finish shall be in accordance with the highest grade practice used in manufacturing military weapons. The quality of workmanship shall not adversely affect safety, function, performance, serviceability, interchangeability and appearance. Finished items and parts shall not exhibit poor materiel and processing such as seams, laps, laminations, ,cracks, visible steps, sharp edges, nicks, scratches, burrs, deformations, missing operations, improper assembly, missing parts or any nonconformance that may affect the quality of the item. Fins and other extraneous metal shall be removed from cast or forged parts. Hammering to shape, salvage operations (including repair by welding) or other similar practices shall not be permitted without prior approval of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-W-63150 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. Unless otherwise stated in the contract or purchase order, the contractor shall submit a first article sample consisting of three recuperator assemblies for evaluation in accordance with provisions of 4.3.2. The contractor shall also submit his approved process control plan for weld overlay procedures with the first article submission (see 3.2.5.3).

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings. The First Article items will be subjected to gymnastication (see 4.5.1), functional firing (see 4.5.2) and testing to verify operating temperature range (see 4.5.6) as a minimum. A change in design or of vendors of any of the dynamic seals in the recuperator shall require that they be qualified in accordance with the provisions set forth in 4.5.7 and that a first article be submitted.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection

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lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, variables sampling or other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.9). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provision of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.1 Recuperators. Each recuperator shall be gymnasticated in accordance with the requirements of 3.3.2 as well as examined for compliance with the requirements of sections 3.2, 3.3.5, 3.5 and 3.6. One in ten recuperators shall be functionally fired in accordance with the requirements of 3.3.3. Functionally fired recuperators shall be examined for compliance with 3.3.4 and 3.3.5.

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4.4.2.2 Parts and assemblies. The parts and assemblies comprising the recuperator assembly shall be inspected in accordance with the requirements of the appropriate drawings and Quality Assurance Provisions (QAPs) pertaining to them.

4.4.3 Requalification inspection of critical components. Requalification inspection shall be performed on critical components when:

a. a new supplier is utilized

b. Significant changes are made in the manufacturing process or item material

c. A design change is made and the potential impact of the change is deemed significant enough by the Government to warrant testing.

4.4.3.1 Critical components. Critical components of the recuperator are those that, if defective, will affect safety and or prevent the recuperator from complying with its functional requirements.

4.4.4 Testing.

4.4.4.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete recuperator assembly. If test requirements cited herein are not met, acceptance of the recuperator or recuperator lot shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study to include a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.

b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.

c. Examine recuperators, partially assembled recuperators and components (including components and subassemblies in process or final assembly) to ensure that materiel containing the same defect is purged from inventory and is not presented to the Government for acceptance.

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d. Submit the result of the failure analysis and the corrective action taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned recuperators for retest.

4.4.5 Inspection equipment. The inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078.

4.4.5.1 Accuracy and reliability. All inspection equipment utilized in the production of the recuperator assembly shall be capable of accurate and repeatable measurements when utilized by various experienced personnel.

4.4.5.2 Contractor responsibility. Unless otherwise specified, the contractor shall be responsible for the acquisition, calibration, maintenance and disposition of all acceptance inspection equipment; which shall comply with MIL-I-45607. The contractor shall also be responsible for providing all equipment calibration procedures developed in accordance with MIL-STD-45662.

4.4.5.3 Equipment utilization. Use of contractor's inspection equipment, when desired by the Government representative, shall be permitted without charge or additional cost to the Government.

4.4.5.4 Government responsibility. Government designed inspection equipment, when required, shall be as specified in the contract or referenced in the technical documentation.

4.4.5.5 Visual inspection standards. Inspection standards shall be utilized for those characteristics requiring visual determination of established criteria. Visual comparison standards shall be subject to the approval of the responsible Government technical agency and shall be of mutual agreement between the contractor and the Government. Each comparison or inspection standard shall be kept under the control of the contractor's inspection element and be positively identified as to the characteristic or condition the standard represents, date established as a standard, control number and identity of contractor and government personnel stabilizing the standard.

4.4.5.6 Government approval. The Government reserves the right to inspect the contractor's inspection equipment and determine availability of equipment, correct utilization, accuracy and

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precision, and that the instruments are of proper type and range to make measurements of proper accuracy.

4.4.6 Inspection of packaging. Unless otherwise specified (see 6.2), inspection to determine compliance with preservation, packaging and marking requirements of the applicable packaging documentation, for the level specified in the contract, shall be as specified in the Quality Assurance Provisions (QAPs) (see 5.9 of MIL-STD-2073-1).

4.5 Methods of inspection. Each recuperator assembly shall be examined for proper assembly, oil level and nitrogen leakage prior to subjecting the recuperator assembly to gymnastication and functional firing.

4.5.1 Gymnastication.

a. The recuperator assembly shall be placed in an appropriate fixture simulating its installation on a complete howitzer, approved by the Government, for gymnastication.

b. The recuperator assembly shall be charged with nitrogen to a pressure of 750+/-50 psi.

c. The recoiling mass shall be pulled 44+/-0.5 inches out of battery and allowed to counter-recoil freely.

d. Step c will be repeated for a total of 20 repetitions.

e. On last extension, all exposed sliding surfaces shall be inspected for scratches, galling and bronze pickup.

4.5.2 Functional firing. Each recuperator assembly selected in accordance with 4.4.2.1 and properly mounted to an M119A1 howitzer shall be functionally fired in accordance with the firing schedule of Table I. Visual examination for weld failures, deformations and any evidence of damage shall be conducted on each recuperator assembly at the conclusion of functional firing. Failure to comply with the requirements of 3.3.3 shall be cause for rejection.

4.5.2.1 Simulation. The contractor may verify the requirements of 3.3.3 using simulation in lieu of functional firing. If simulation is utilized the contractor shall submit his simulation plan and test report to the government for approval.

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4.5.3 Contaminated oil test. Immediately after functional firing, fully depress the ordnance (-100 mil), remove a minimum of two ounces of oil from the recuperator into a straight sided glass beaker. Metallic and non-metallic particles allowable in the oil sample shall not exceed 400 microns in size.

4.5.4 Aerated oil test. Immediately after removal of oil sample for 4.5.3, elevate the ordnance to 180 +10 mil. Allow a period of five minutes to lapse. Slowly withdraw a minimum of four ounces of oil from the recuperator into a straight sided glass beaker. Within the first minute after withdrawal, the oil shall be five percent or less foam by volume. When exposed to the atmosphere for fifteen minutes, the aeration bubbles shall disappear.

4.5.5 Leak tests.

4.5.5.1 Nitrogen leak test. During and immediately after the completion of functional firing or gymnastication, the nitrogen leak test shall be conducted as follows:

a. The recuperator assembly shall return the ordnance completely into battery. If the ordnance does not return to battery completely, check the nitrogen pressure in the recuperator. Loss of pressure over that measured prior to gymnastication/functional firing, within the accuracy of the measuring equipment, shall be cause to reject the recuperator.

b. The recuperator assembly shall be examined for compliance with the requirements of 3.3.5.1. The recuperator shall be left pressurized for a period of 24 hours. After compensation for temperature change and measurement error, there shall be no loss of pressure. Alternate test methods may be submitted to the Government for approval.

4.5.5.2 Oil leak test. During and immediately after the gymnastication/functional firing test, the leak test shall be conducted as follows:

a. During test, the recuperator shall be visually checked for excessive leakage. Leakage in excess of 15 drops per minute shall be cause for rejection of the assembly.

b. After withdrawal of the oil sample for 4.5.3, allow for a period of ten minutes to lapse, then visually check for oil leakage from the front and rear of the recuperator. Leakage shall not be more than 15 drops per minute.

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4.5.6 Operating temperature range verification. Each of the first article samples shall be fired at the temperature extremes, specified in 3.3.6, in accordance with Table II.

Table II: Operating Temperature Verification

Round Number 1/	Prop Charge	Zone	Quadrant Elevation (mil)	Recoil Length (inches) 2/
		-50 degrees F		
1	M67	7	90	38/43
2	M67	7	90	38/43
3	M67	7	90	38/43
4	M67	7	90	38/43
5	M67	7	90	38/43
		+145 degrees F		
6	M67	7	90	38/43
7	M67	7	90	38/43
8	M67	7	90	38/43
9	M67	7	90	38/43
10	M67	7	90	38/43

1/ Standard 105mm projectile inert loaded to 30.85 ± 0.35 pounds.

2/ Recoil lengths are for information only.

3/ The assembly shall be stabilized at each temperature prior to firing the assigned group.

4.5.7 Dynamic seal qualification. In order to qualify, a dynamic seal must withstand firing a five round group on three separate occasions each at each temperature extreme. The seal must also complete a durability test of 3000 M67 zone 7 charges fired at 200 mils Quadrant Elevation with no seal failures or 6300 M67 zone 7 charges fired at a QE of 200 mil with one seal failure. The Rock Island Arsenal hydraulic simulator may be used to qualify dynamic seals only.

5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of a complete recuperator assembly preserved for the level of protection specified in the contract (see 6.3).

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5.2 Levels A, B and C. Preservation, packaging and marking shall be performed for the level of protection specified in the contract.

5.3 Repair parts. Repair parts shall be preserved, packaged and marked in accordance with special packaging instructions or other packaging requirements as specified in the contract (see 6.3).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The recuperator assembly is one of two major components of the recoil mechanism utilized in the Howitzer, Light, Towed, 105mm: M119.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Assembly area cleanliness. Recuperator (Recoil mechanism) assembly areas should be air conditioned using a unit with a 50 micron filter. The area temperature should be maintained between 68 and 72 degrees Fahrenheit. The air flow rate should maintain the area air pressure between .75 and 1.5 inches of water. Relative humidity should not exceed 50 percent. Assembly areas should be clean and free of dust, lint and abrasive materiels that may become entrapped in assembled mechanisms. No grinding, filing, scraping or

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any other operation producing abrasive dust should be performed in the assembly area.

6.4 Firing records. One copy of each final Proving Ground Firing Record for each recuperator tested shall be sent to AMCCOM, ATTN: SMCAR-QAR-Q and ATTN: SMCAR-FSA-T, Picatinny Arsenal, NJ 07806-5000 for informational purposes.

6.5 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAR-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.6 Data requirements.

Reference paragraph	Number	Title
4.5.2.1	DI-NDTI-80603	Test Procedure
4.5.2.1	DI-NDTI-80604	Test Report

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.8 Subject term (key word) listing.

Towed weapon
Light armament

6.9 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval as directed by the contracting activity.

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6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1015-0339)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-71172A (AR)	940509

3. DOCUMENT TITLE: CYLINDER, RECUPERATOR, RECOIL, 105mm HOWITZER

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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